

MATH 55 FINAL EXAM, PROF. SRIVASTAVA
FRIDAY, MAY 13, 2016, 7:10–10:00PM, 100 LEWIS HALL.

1. Circle true (**T**) or false (**F**) for each of the following. There is no need to provide an explanation.

(a) (2 points) The compound proposition

$$(p \rightarrow (q \wedge \neg q)) \rightarrow \neg p$$

is a tautology.

T F

(b) (2 points) There exist integers x and y such that $21x + 54y = 1$.

T F

(c) (2 points) A graph is 3-colorable if and only if all of the circuits in it have length divisible by 3.

T F

(d) (2 points) If E and F are events in a probability space such that $p(E) = 0.99$ and $p(F) = 0.99$ then E and F cannot be independent.

T F

(e) (2 points) Let I_n denote the number of injective functions from $\{1, 2, 3, \dots, n\}$ to $\{1, 2, \dots, 55\}$. If $m \geq n$ then it must be the case that $I_m \geq I_n$.

T F

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2. (10 points) Suppose A is a (possibly infinite) set. Prove that A and its power set $\mathcal{P}(A)$ cannot have the same cardinality. (hint: Prove that there is no bijection $f : A \rightarrow \mathcal{P}(A)$ by considering for any function f the subset $B = \{x \in A : x \notin f(x)\}$.)

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3. Consider the following one player game. The game begins with n shapes, each of which is either a circle or a square. In each move, I have two options: (A) choose two shapes of the same type and replace them by a square, OR (B) choose two shapes of different types and replace them by a circle. The game is over when there is exactly one shape left, and I win if that shape is a circle (otherwise I lose).

(a) (4 points) Prove by induction that the game always terminates in a finite number of moves (regardless of the initial state).

(b) (6 points) Prove that I win the game if and only if the initial number of circles is odd.

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5. (6 points) Calculate the remainder $(-55)^{2016} \bmod 13$.

6. (6 points) Suppose $k \geq 1$ and $(x_1, \dots, x_k)$ is a randomly chosen k -permutation of $\{1, \dots, n\}$ (i.e., an ordered arrangement of k distinct elements, chosen uniformly from all such arrangements). What is the probability that it is a strictly increasing sequence, i.e., that

$$x_1 < x_2 < \dots < x_k?$$

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7. (10 points) Consider a dial having a pointer that is equally likely to point to each of n regions numbered $1, 2, \dots, n$. If I spin the dial 3 times, what is the probability that the sum of the selected numbers is exactly n ?

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8. Suppose I roll a standard 6-faced die 7 times. Let X be a random variable equal to the number of **distinct** faces that appear (a face is a number in $\{1, 2, 3, 4, 5, 6\}$.)
- (a) (6 points) What is the expectation of X ?

- (b) (4 points) What is the variance of X ?

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10. Prove or disprove:

- (a) (5 points) There exists a simple graph with 6 vertices and 5 edges such that G has an Euler circuit.

- (b) (5 points) There exists a simple graph G with 6 vertices and 7 edges such that G has an Euler circuit.

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11. (10 points) Prove by induction on k that every simple graph with n vertices and k edges has at least $n - k$ connected components.